



“Unlocking More Possibilities for Tea Processing.”

- **Area:**

Industrial Equipment, Arduino, Corporate Partnerships with ***Shannai Tea Factory***

- **Feature:**

Experimentation, industrial factories

- **Duration:**

May.2024 - Jul.2024

- **Team project:**

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We made joint efforts in previous research, model making, and I finished functional framework and final layout on my own



Project Background

Withering is a crucial step in tea processing. It allows the tea leaves to lose moisture, shrink in volume, soften in texture, and enhances enzyme activity, significantly improving the tea's quality. This process also imparts a unique flavor to the tea and extends its shelf life.

During withering, even slight changes in various factors can lead to significant differences in flavor. Therefore, withering is a process that requires precise control and strict supervision.



During our preliminary research, we established contact with Shannai Tea Factory in Chaozhou and learned that the factory still relies on traditional manual operations. Professional technicians and workers handle the entire tea planting and production process by hand.

In particular, during the tea withering process, due to its short duration, large workload, and high level of specialization, typical manual operations are often insufficient to meet production demands.

Based on this, we aim to design an outdoor withering assistive tool, distinct from the existing indoor withering troughs. By mechanically modifying the withering trough, we seek to enable real-time monitoring of tea leaf conditions, implement corresponding measures, facilitate even spreading, and provide timely protection for the tea leaves in the event of unexpected rainfall. This will offer a new approach for the mechanization of tea production.

PEST Analysis

The Chinese tea market boasts a rich cultural heritage and vast development potential. Government support, economic growth, changing consumer habits, and technological advancements all provide strong backing for the market's growth. However, competition is fierce, and brand building and innovation are key to success.

P POLICY

Government support for tea industry, agricultural subsidies, and trade policies influence the adoption of intelligent withering troughs.

Environmental regulations drive higher design and production standards for sustainable technologies.

E ECONOMY

Economic growth boosts consumer purchasing power, increasing demand for high-quality and premium tea products, driving innovation in production methods.

Intense market competition forces tea companies to adopt cost-effective technologies like intelligent withering troughs to stay competitive.

S SOCIETY

As consumer preferences shift toward healthier and more sustainable products, demand for teas produced with advanced technologies rises.

Tea's cultural significance encourages companies to maintain traditional methods while integrating modern innovations to meet evolving consumer expectations.

T TECHONOLOGY

Advancements in automation, AI, and IoT help optimize tea production processes, making withering more efficient and consistent.

The use of data analytics allows for precise control over the tea withering process, ensuring higher quality and uniformity in production.

Research and Interview Findings

Withering Duration for Tea Leaves

The withering time depends on the temperature of the day; the higher the temperature, the shorter the time required. It is essential to have manual monitoring, and if it starts to rain unexpectedly, the tea leaves must be quickly moved indoors to avoid affecting the quality.

Unmet Needs in the Withering Process

There is a need for machine assistance in spreading and turning the tea leaves, as well as intelligent temperature monitoring and cooling systems.

Current Use of Machinery in the Withering Process

Machinery is primarily used for indoor withering troughs for roasting, typically only when the weather is bad. Tea leaves processed with mechanical roasting don't have the same quality. Most often, manual sun drying is preferred, as sun withering produces better quality tea.

Types of Tea in the Factory with High Demand for Sun Withering

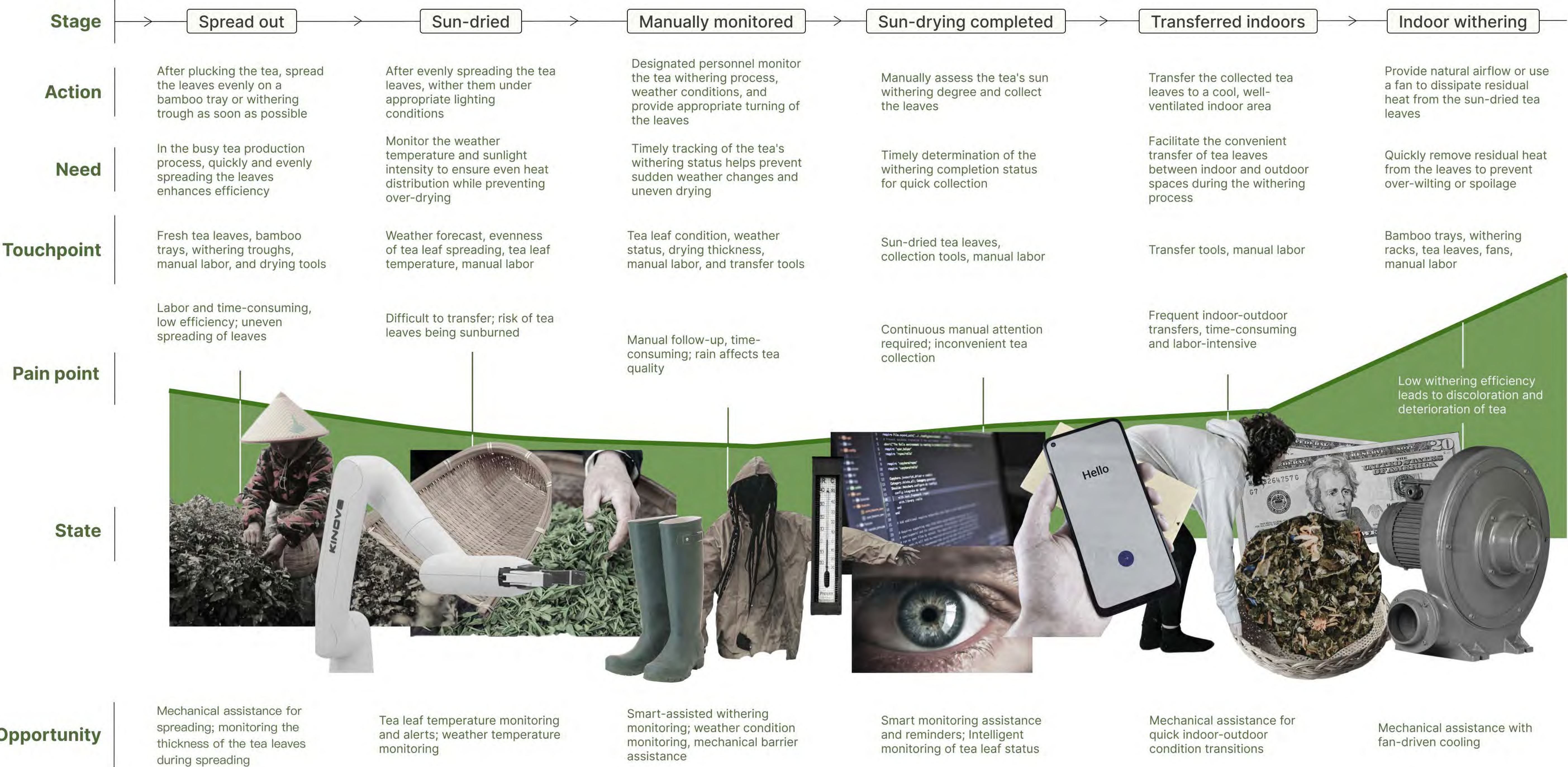
Oolong tea, in particular, requires precise control over the withering process to reach the semi-fermentation stage and bring out the delicate floral aroma. Sun withering is essential to achieve high-quality Oolong tea.

Key Considerations for Outdoor Sun Withering

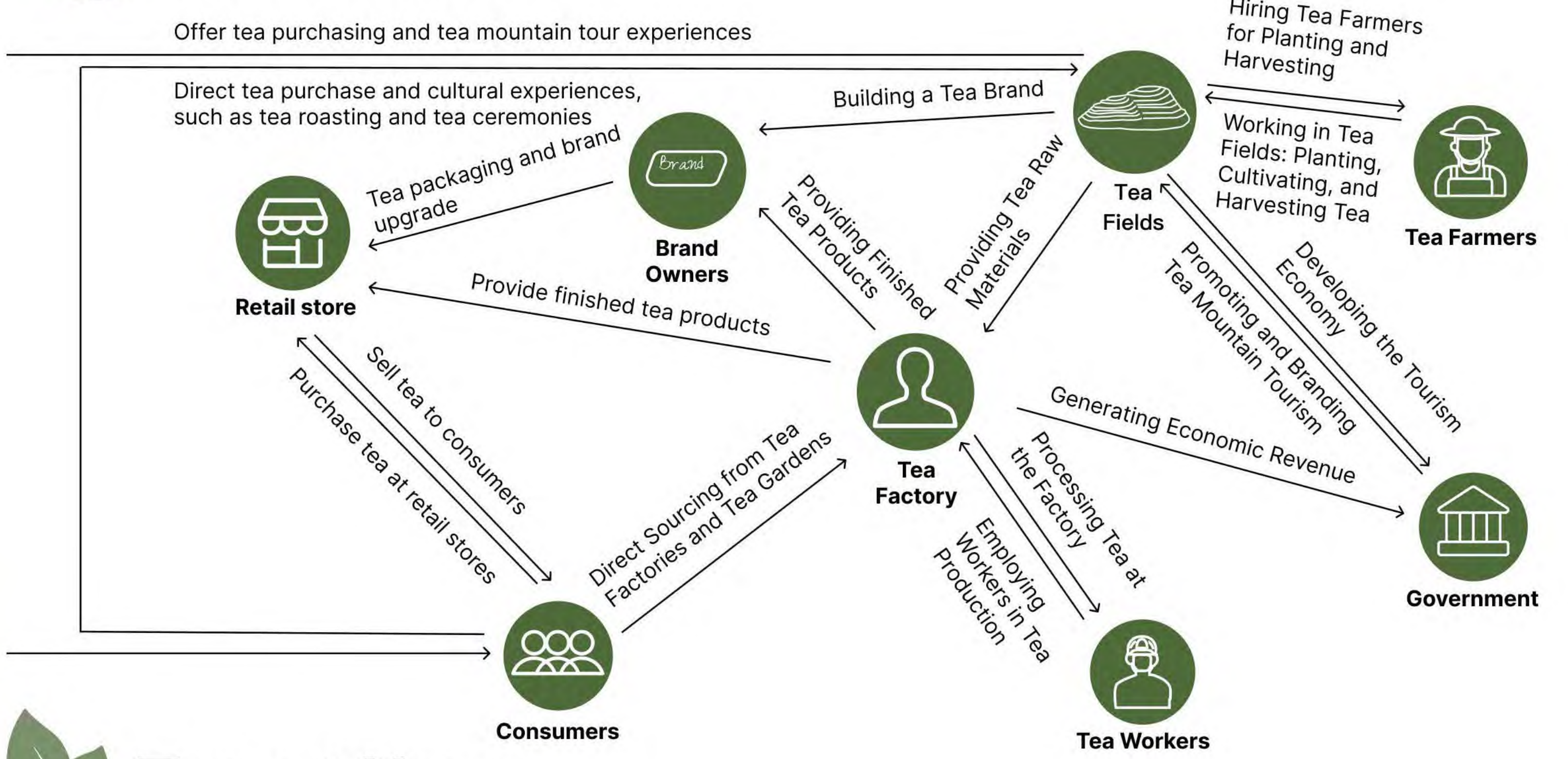
If the temperature exceeds 30°C on a given day, we generally do not dry the tea between 11 AM and 3 PM. The overall tea-making process is very busy and tight. Once the tea leaves are picked, they need to be quickly withered to prevent overheating and fermentation, which can cause them to turn red. When spreading the tea leaves, it is crucial to lay them out evenly. For thin layers, no turning is necessary, but thicker layers usually require 2-3 manual turns.



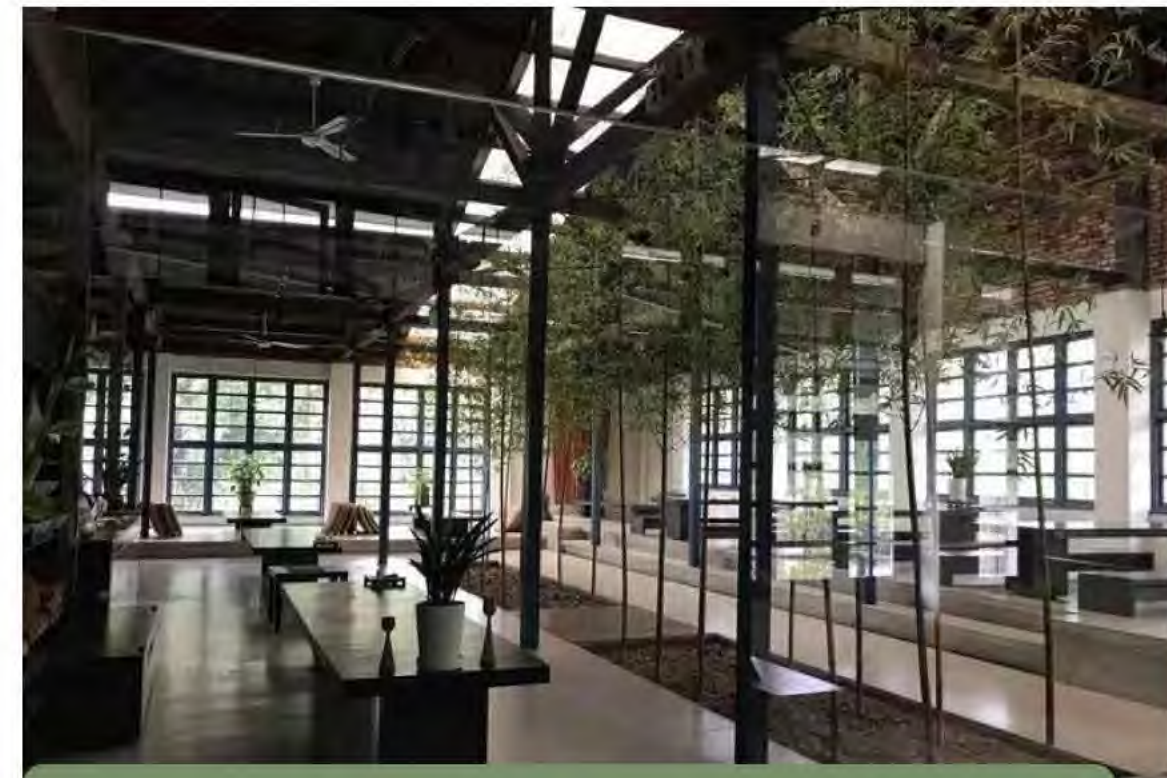
User Journey Map



Stakeholders



Target Users



User Name: Shannai Tea Factory
Scale: Small to Medium-sized Enterprise
Main Business: Alcohol, Beverage, and Processed Tea Manufacturing

Operating Model:
Traditional manual operations, with professional technicians and workers completing the entire tea planting and production process by hand. In the tea withering process, due to its short duration, high workload, and specialized nature, typical manual operations struggle to meet production demands.

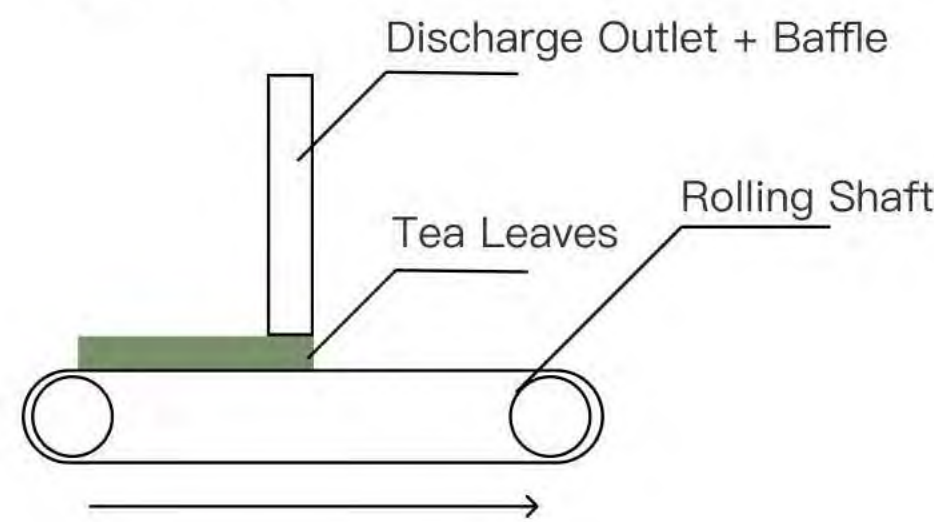
- Pain Points:**
- Small to medium-sized tea factory with a limited commercial scope and low budget.
 - Lack of skilled technical workers.
 - High workload during withering, leading to increased pressure on workers.
 - Located in the southeastern region, where rapid weather changes can easily disrupt the tea withering process.

Customer Needs:
Leverage advanced scientific technology to replace traditional manual production methods, achieving progress in production techniques and cost savings. This will enable the modernization and automation of the tea withering process.



Functional Framework

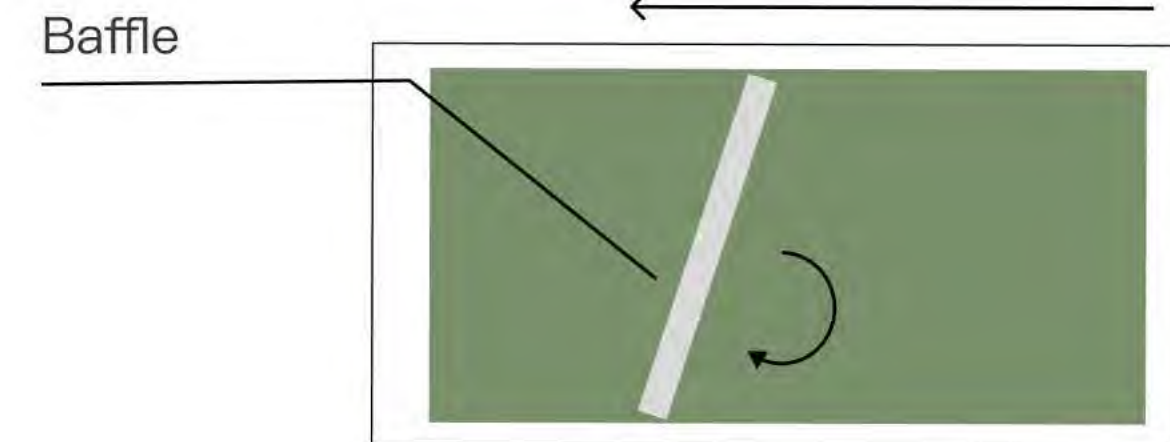
Spreading



Spreading is uniform, eliminating the need for manual labor

Tea leaves are transported through the discharge outlet, with the baffle controlling the spreading thickness, ensuring an even spread. The rolling mechanism evenly distributes the tea leaves.

Turning



Automatic flipping ensures uniform wilting

The baffle rotates and moves simultaneously, flipping the tea leaves to ensure even wilting.

Functions

Monitoring

Judgment

Execution

Real-time weather monitoring with automatic shielding when rain is detected to prevent wetting

Machine-assisted tea leaf assessment, reducing labor costs and alerting to wilting conditions

Monitoring environmental changes such as temperature and humidity, with real-time feedback

Turning tea leaves to ensure even wilting

Monitoring temperature rise due to sunlight and issuing alerts before overexposure occurs

Machine-assisted tea leaf assessment, reducing labor costs and alerting to wilting conditions

Rainwater Monitoring

Moisture content is measured (weighing and recording before and after wilting, with weight reduction indicating moisture change)

Real-time Temperature Monitoring

Real-time Humidity Monitoring

Real-time Wind Speed Monitoring

Monitor the temperature difference between the top and bottom layers

Monitor the tea leaf temperature

Visual Recognition Multispectral/Hyperspectral Camera: Captures multi-band images of the tea leaves. Image Analysis Software: Processes image data and provides visual charts of the tea leaf wilting status

When rainfall reaches x milliliters, conditions are met

The percentage of water loss is calculated to determine if it is within the normal range

If wind speed falls below a certain threshold

If the temperature difference exceeds a certain value, it is considered uneven

If the temperature exceeds a certain threshold (X value)

Compare features such as [Hue Mean, Hue Standard Deviation, Green Body Mean] with the standard model

The automatic cover system activates

Weather data is uploaded to the backend, and a notification is triggered

A reminder is issued

Data is uploaded to the backend

The blower system activates

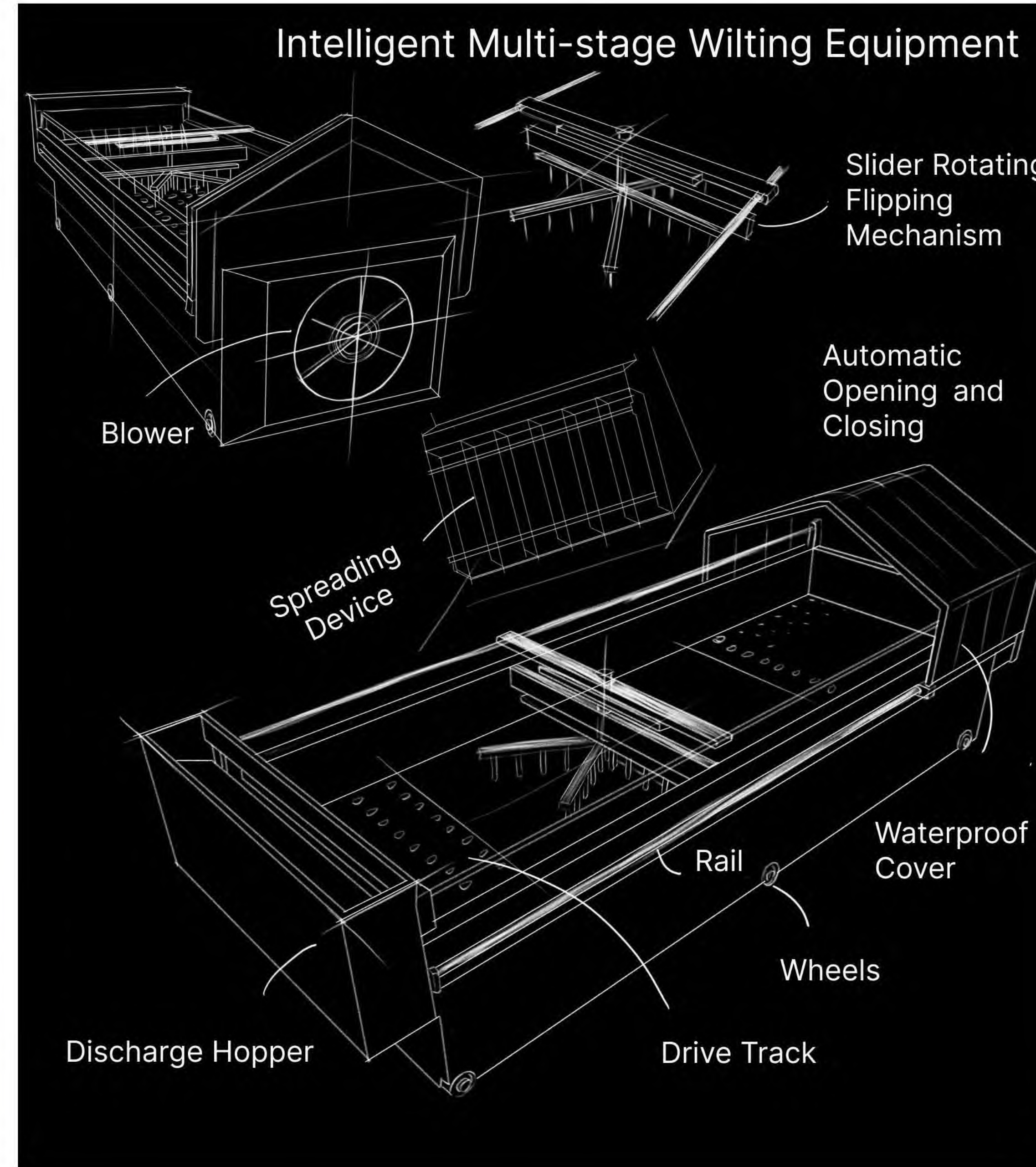
The flipping mechanism activates

The cover is closed to provide shade

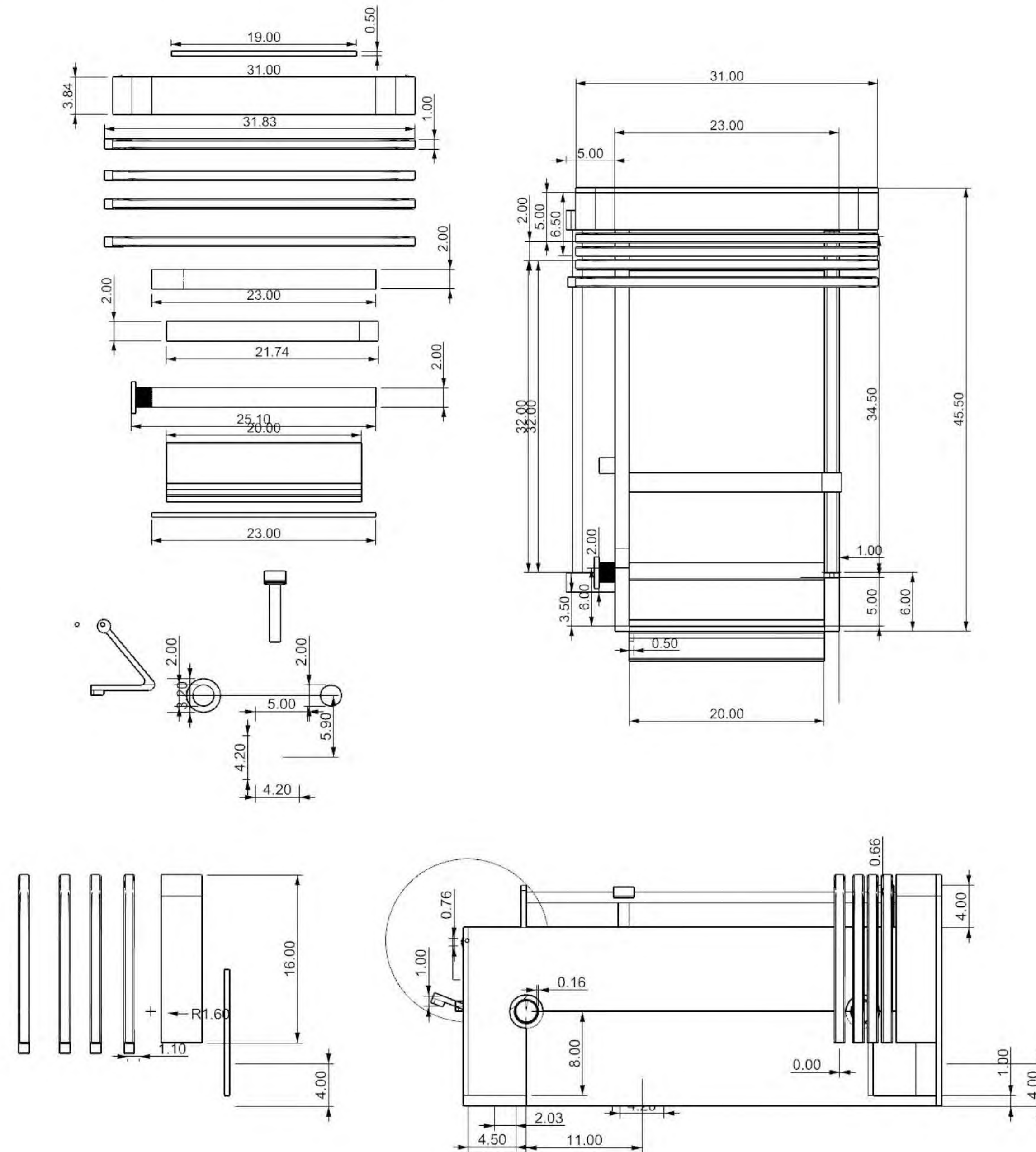
If the comparison is successful, a notification is triggered



Design Sketches



Design Components and Dimensions





Code

```

#include "HX711.h"
#include <Servo.h>
#include <OneWire.h>
#include <DallasTemperature.h>

#define DHTTYPE DHT11
#define DHTPIN 4
#define ONE_WIRE_BUS A0

float Temperature = 1.0;
float Humidity = 1.0;
float temp1=0;
float temp2=0;
float rain=0;
float weight=0;
int time = 0;
int tempgap = 0;

int z;
int r=0;
int t=0;

void sendmsg() {
  mySerial.print("h");
  mySerial.println(Humidity);
  mySerial.println("t");
  mySerial.println(Temperature);
  mySerial.println("a");
  mySerial.println(temp1);
  mySerial.println("s");
  mySerial.println(temp2);
}

void sendmsg0() {
  mySerial.print("h");
  mySerial.println(Humidity);
  mySerial.println("t");
  mySerial.println(Temperature);
  mySerial.println("a");
  mySerial.println(temp1);
  mySerial.println("s");
  mySerial.println(temp2);
}

void setup() {
  mySerial.begin(9600);
  pinMode(DHTPIN, OUTPUT);
  pinMode(ONE_WIRE_BUS, INPUT);
  digitalWrite(DHTPIN, HIGH);
  delayMicroseconds(100);
  digitalWrite(DHTPIN, LOW);
  delayMicroseconds(100);
}

void loop() {
  if (mySerial.available()) {
    String serialData = mySerial.readStringUntil('\n');
    for (int i = 0; i < serialData.length(); i++) {
      str[i] = serialData[i];
    }
    sscanf(str, "%f %f %f %f %f %f", &values[0], &values[1], &values[2], &values[3], &values[4], &values[5]);
    Number1.print(values[0]);
    Number2.print(values[1]);
    Number3.print(values[2]);
    Number4.print(values[3]);
    Number5.print(values[4]);
    Number6.print(values[5]);
  }

  if (mySerial.available()) {
    String serialData = mySerial.readStringUntil('\n');
    for (int i = 0; i < serialData.length(); i++) {
      //char c = serialData.charAt(i);
      //numbers[i] = c - '0'; // 将字符转换为整数
      numbers[i] = serialData.charAt(i) - 48;
    }
    for (int i = 0; i < 200; i++) {
      digitalWrite(step1Pin, numbers[0]);
      digitalWrite(step2Pin, numbers[1]);
      digitalWrite(step3Pin, numbers[2]);
      digitalWrite(step4Pin, numbers[3]);
      digitalWrite(step5Pin, numbers[4]);
      digitalWrite(step6Pin, numbers[5]);
      delayMicroseconds(1000);
    }
    if (digitalRead(9) == 1 && r == 0) {
      int bt3=0;
      int bt4=0;
      int bt5=0;
      int bt6=0;
      int bt7=0;
      int bt8=0;
      int bt9=0;
      int bt10=0;
      int bt11=0;
      int i=1;
      char auth[]="bb52842ca4ee"; //自己的设备码
      char ssid[]="1234";
      char passwd[]="12345678";
      BlinkyNumber Number1("num-h"); //温度
      BlinkyNumber Number2("num-t"); //湿度
      BlinkyNumber Number3("num-a"); //雨量
      BlinkyNumber Number4("num-s"); //重量
      BlinkyNumber Number5("num-r"); //时间
      BlinkyNumber Number6("num-h"); //雨量
    }
  }
}

```



Model Making



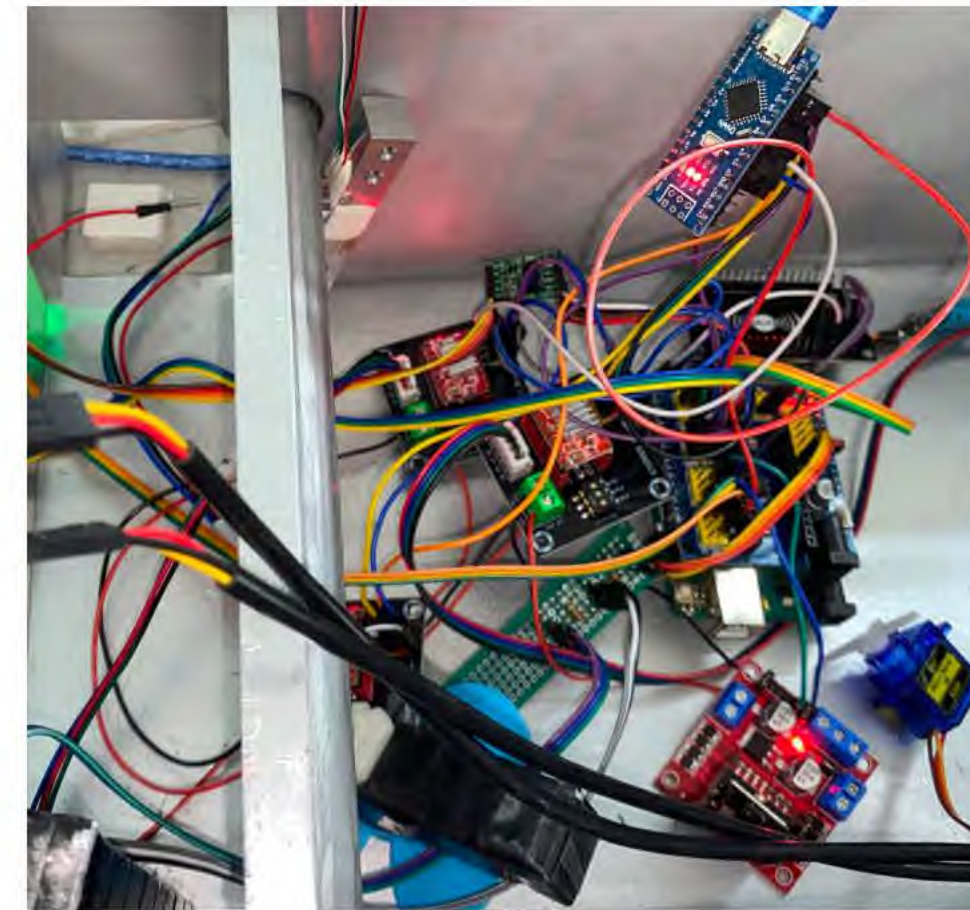
3D Model Construction

Assemble the model by sanding the 3D printed parts with sandpaper until smooth



Model Painting

Apply chrome-plated spray paint with a mirror finish to the model's surface to enhance its texture



Wiring Connections

Use Arduino to connect the wiring and secure or extend it based on the positions of the model's components



Module Assembly

Use Arduino to connect the wiring and secure or extend it based on the positions of the model's components



Model Functionality and Module Display



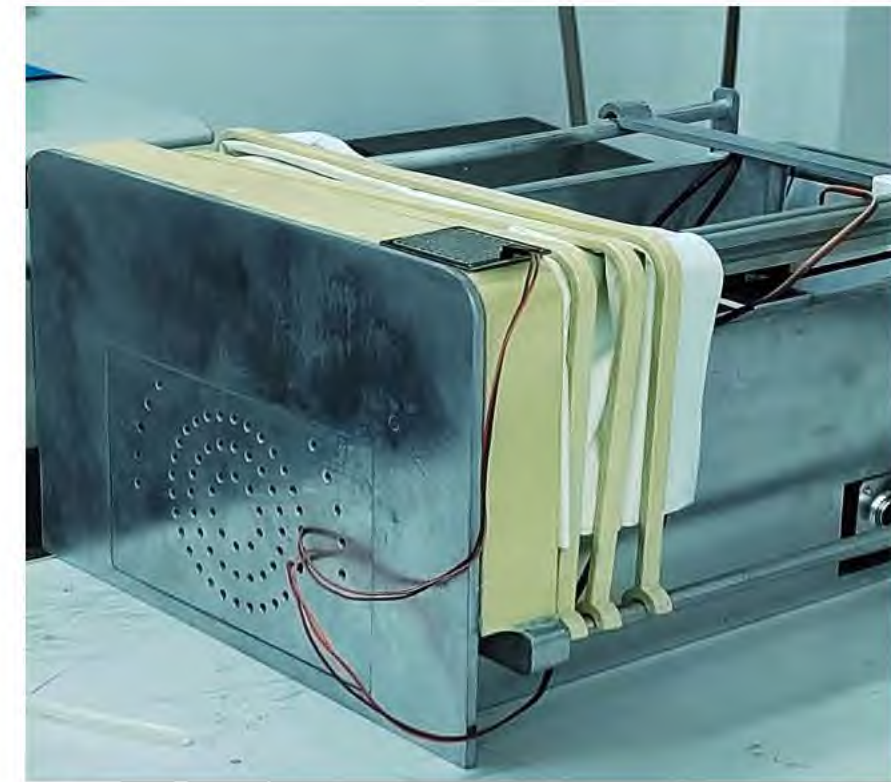
Tracks

Evenly spread the tea leaves concentrated in the withering trough. The 42-stepper motor drives the rotation of the tracks in reverse



Withering Shed

Composed of a 42-stepper motor, waterproof and light-blocking custom fabric, and support rods. It receives signals from the rainwater monitoring system and is driven by the 42-stepper motor to open or close



Blower System

Consists of a fan and fan drive unit, assisting with ventilation in the withering trough



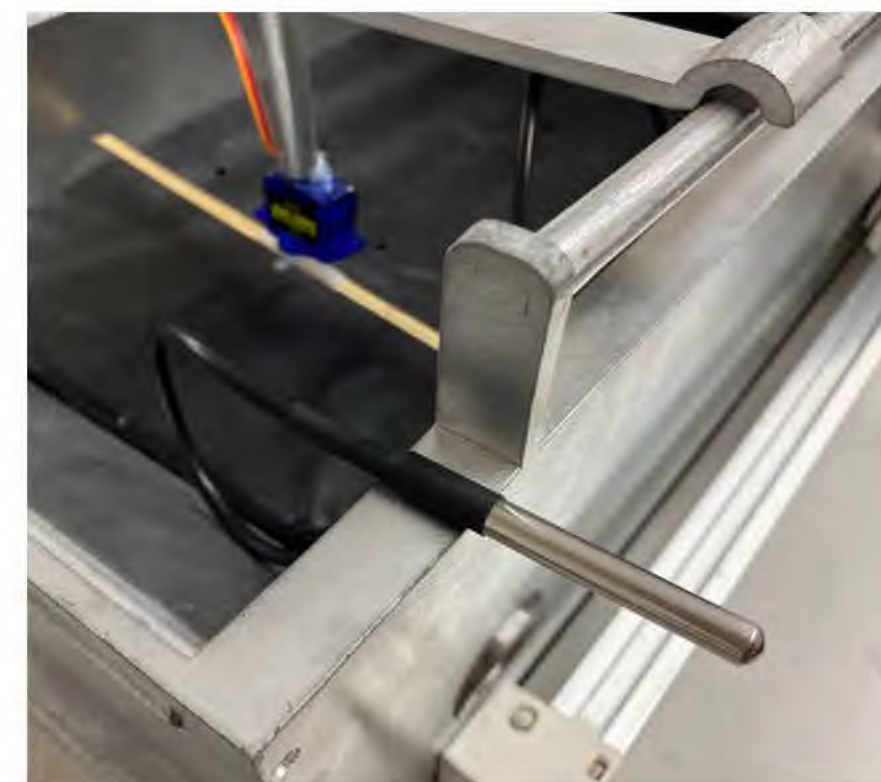
Prototyping and Presentation



Project Feedback



—The person in charge of each function in the project team reports on the completion status of their tasks in this project



DS18B20 Temperature Sensor

Monitors temperature differences in different areas of the tea leaves, determining when to turn the leaves and transmitting the signals to the app



SG90 9g 360° Servo Motor

Used to evenly spread and flip the tea leaves through movement and rotation



Raindrop Sensor

Detects rain and sends a signal to deploy the withering shed. When no rain is detected, it signals to retract the shed